

ROHIT SARKAR

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TECHNICAL SKILLS

Languages: Java, Python, C/C++, JavaScript, TypeScript, HTML, CSS

AI / ML: Spring AI, ML model integration, fraud detection, anomaly scoring, NLP concepts, prompt engineering, LLM API integration

Frameworks: Spring Boot, Spring Security, FastAPI, Angular, React, Node.js, Kafka, REST APIs

Databases: PostgreSQL (EXPLAIN ANALYZE, query optimisation, indexing), MySQL, TimescaleDB, MongoDB

Cloud / DevOps: AWS (EC2, S3, RDS, IAM, CloudWatch), Docker, Jenkins, CI/CD, Git, GitHub

Testing: JUnit, Mockito, Postman, Insomnia, SoapUI, Browser DevTools

Concepts: Microservices, Distributed Systems, Event-Driven Architecture, MLOps, OAuth2, HIPAA, System Design, Agile/Scrum, Technical Documentation, API Documentation

EXPERIENCE

Software Engineer I

Bengaluru | Nov 2023 – Present

Deloitte

- Designed and deployed **AI-powered claims validation pipeline** using Spring AI and machine learning models for real-time fraud-risk scoring and anomaly detection, reducing manual review effort by **20%** across enterprise healthcare systems.
- Architected **10+ production microservices and RESTful APIs** integrating AI inference endpoints into enterprise workflows, maintaining **99.9% data integrity** and boosting operational efficiency by **30%**.
- Built **event-driven AI inference pipelines** using Kafka for asynchronous model invocation, confidence score routing, and dead letter queue handling for failed predictions – reducing validation errors by **25%**.
- Deployed and managed AI-integrated services on **AWS (EC2, S3, RDS)** with CI/CD pipelines; enforced least-privilege **IAM** roles, implemented security-by-design, and ensured full **HIPAA compliance** via Spring Security and OAuth2.
- Optimised ML model serving latency by **40%** through database query tuning (PostgreSQL EXPLAIN ANALYZE, composite indexing), schema redesign, and time-series workload migration to **TimescaleDB**.
- Achieved **85%+ code coverage** via 200+ JUnit/Mockito tests; led code reviews, authored **technical and API documentation**, defined AI service boundaries and API contracts, and mentored junior engineers on responsible AI integration patterns.

Technical Content Writer Intern

Remote | Dec 2022 – Jul 2023

GeeksForGeeks

Authored 20+ top-read articles on data structures, algorithms, and system design with complexity analysis – referenced by thousands of FAANG interview candidates globally.

Full Stack Developer Intern

Chennai | Jan 2023 – Mar 2023

Full Creative

Built 3 cross-platform desktop productivity apps using Electron JS, Node.js, TypeScript, and Excel integrations; delivered responsive UI/UX within Agile sprint deadlines.

PROJECTS

Library Management System – Spring Boot, Angular, PostgreSQL, REST

library-management-system-vert-eight.vercel.app

Full-stack library platform with Spring Boot backend, Angular 17 frontend, and PostgreSQL on Neon cloud. Features book and member management, issue/return workflow with overdue detection, soft delete for audit integrity, global exception handling, and Swagger/OpenAPI documentation.

URL Shortener API – FastAPI, Python, Docker, REST

github.com/Rohit-Sarkar55/URL-Shortener

Production-ready Python microservice with RESTful endpoints, Pydantic request validation, input sanitisation, duplicate detection, and expiry logic; containerised with Docker for scalable deployment.

Keeper App – Full Stack Notes Manager – React, Node.js, Vercel

my-keeper-eta.vercel.app

Full-stack CRUD application with React hooks, persistent state management, reusable component architecture, and automated CI/CD deployment via Vercel.

COMPETITIVE PROGRAMMING AND ACHIEVEMENTS

LeetCode Knight – Rating 1951, Top 3% globally | Global Rank 451 (Weekly Contest 329) | 1000+ problems solved across LeetCode, CodeChef, Codeforces, and GeeksForGeeks

Contest Rankings: CodeChef 4-Star (1856) | Codeforces Pupil (1260) | Google Kick Start 2022 Round B – Global Rank 2036 | Meta Hacker Cup 2022 – Qualified Round 2, Rank 2711

EDUCATION

Bachelor of Technology, Computer Science Engineering

2019 – 2023

St. Thomas' College of Engineering and Technology, Kolkata

CGPA: 9.06 / 10.0

CERTIFICATIONS

AI Fluency: Framework and Foundations – Anthropic (2026) | Programming Fundamentals – Duke University, Coursera | Data Structures and Algorithms using Python – NPTEL | MERN Stack Web Development – Udemy